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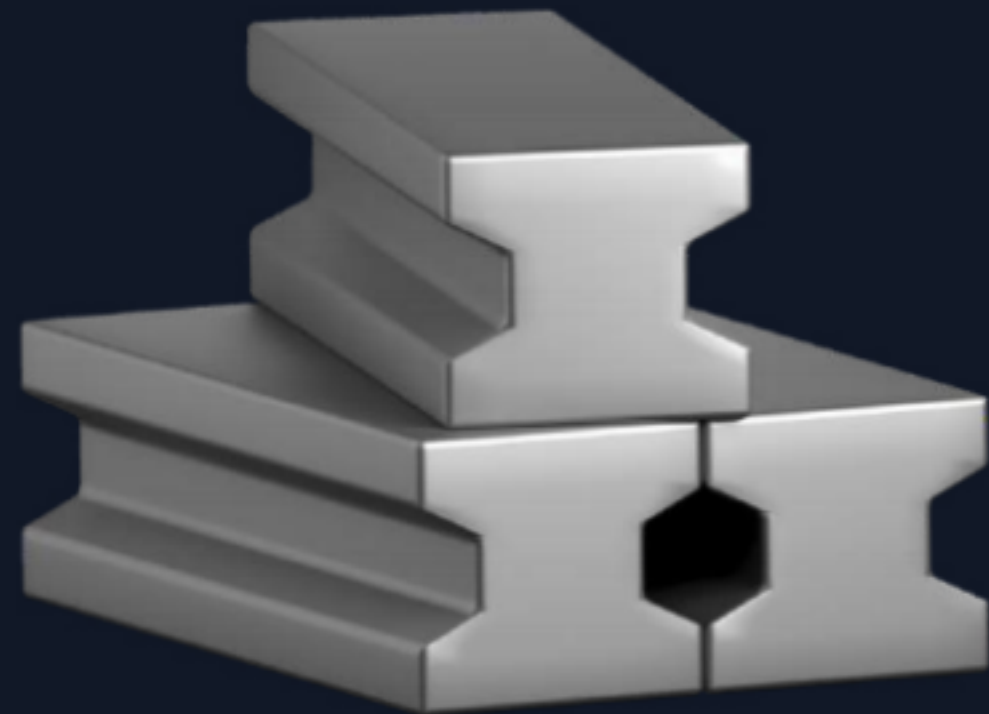
CLASS X - SCIENCE

METALS AND NON-METALS

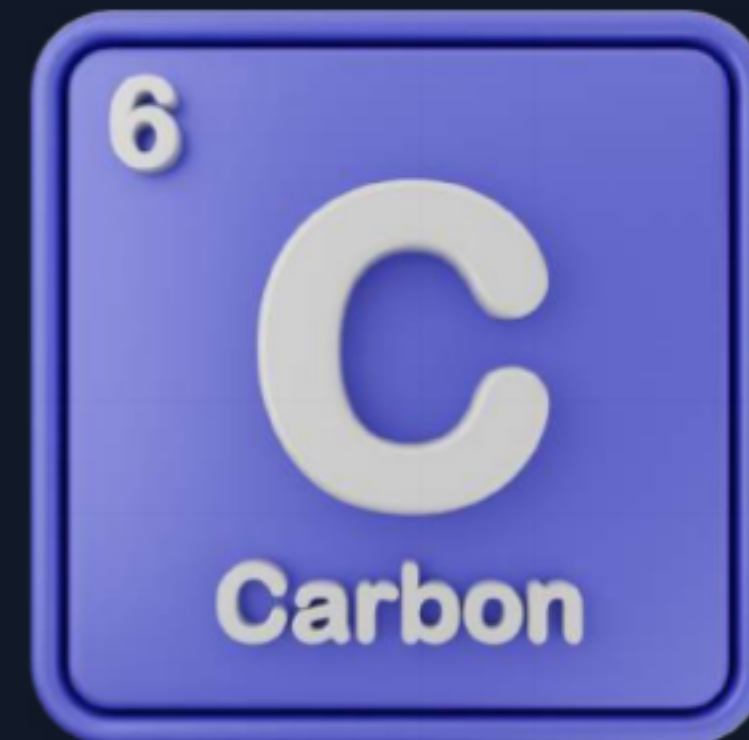
PRASHANT KIRAD

Types of Elements

METALS



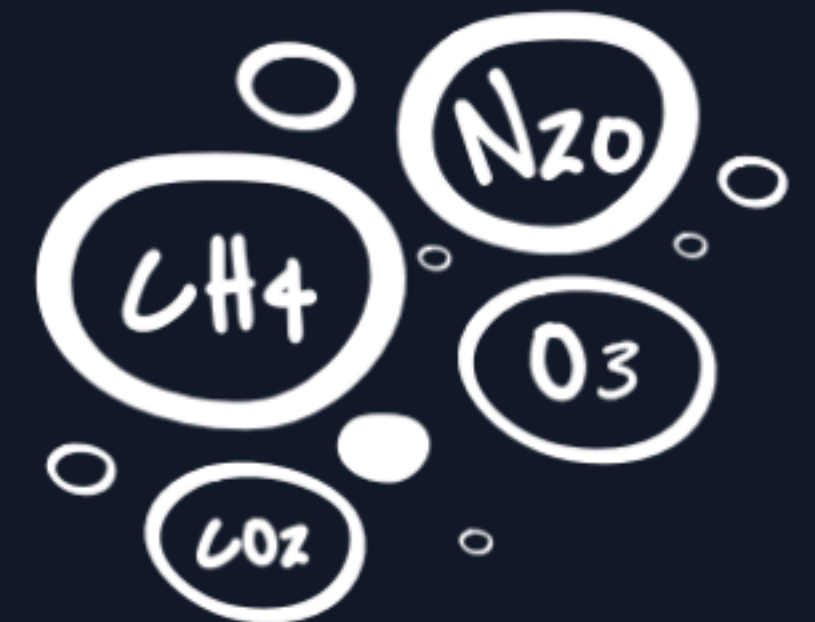
NON METALS



METALLOIDS



NOBLE GAS



Metal

- Metals are elements that are generally hard, lustrous (shiny), malleable, ductile and good conductors of heat & electricity.
- Metals are elements that generally **lose one or more valence electrons** to form positively charged ions (cations).

For example:

- Sodium (Na)*
- Magnesium (Mg)*
- Iron (Fe)*
- Aluminium (Al)*



Na $\rightarrow$ 11

Na⁺

K L M

2 8 (1) $\rightarrow$

(1, 2, 3)

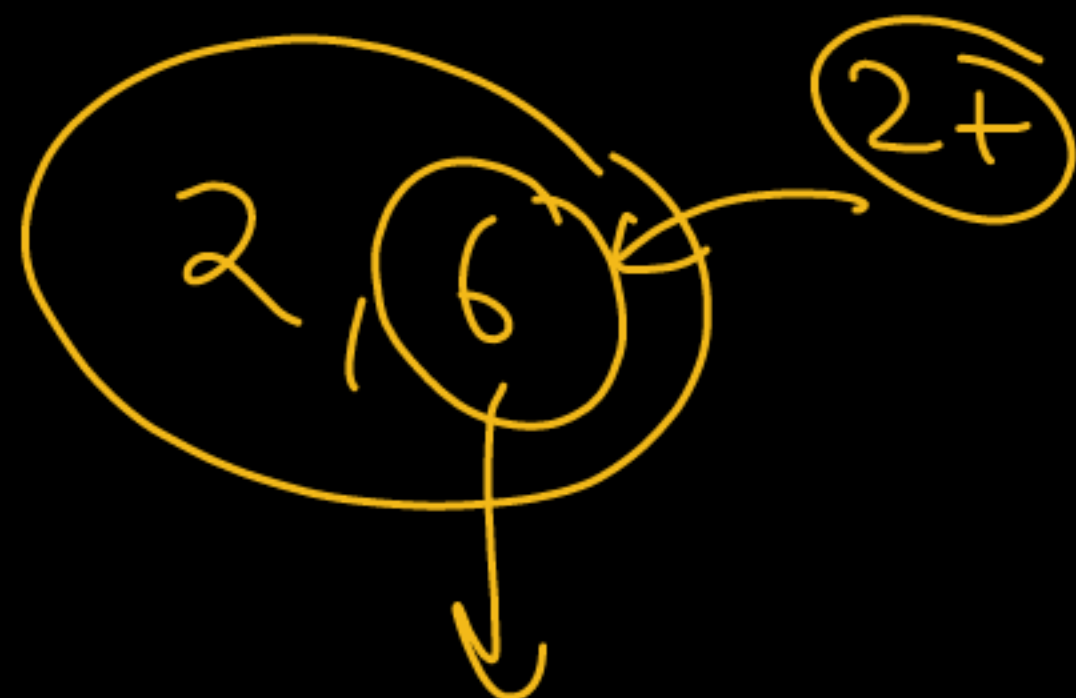


Non-Metals

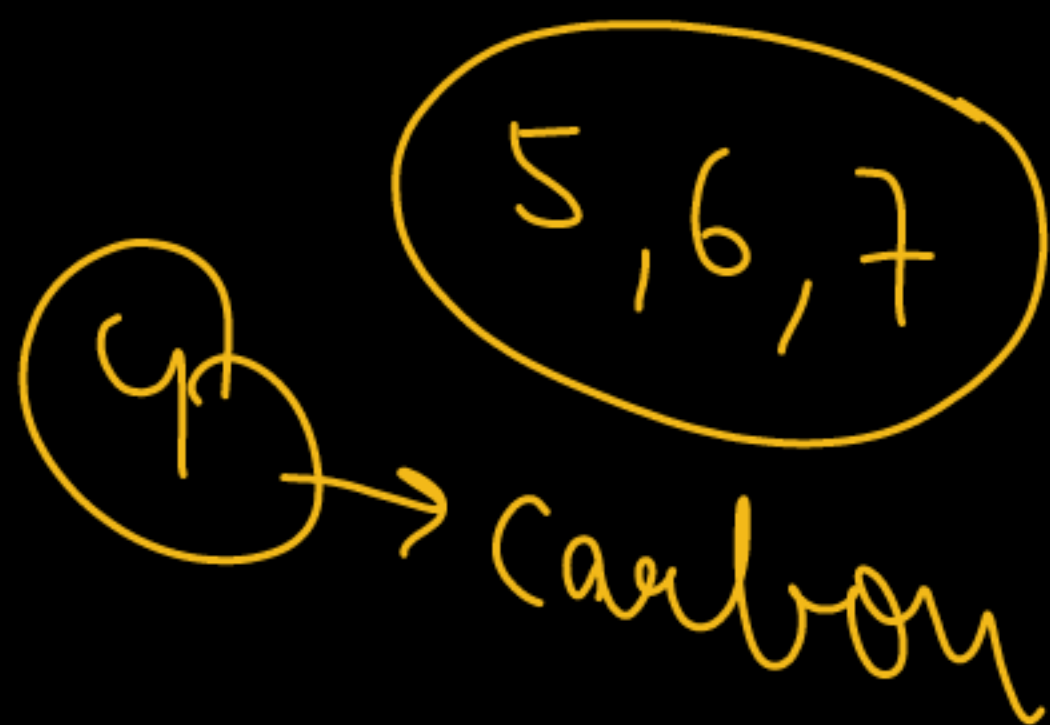
- Non-metals are elements that lack the properties of metals. They are dull, brittle and poor conductors of heat and electricity.
- Non-metals are elements that generally **gain one or more electrons** to form negatively charged ions (anions).

For example: *Hydrogen, Helium, Iodine, Sulfur, Oxygen & carbon*





Non-Metal



Metalloids

Metalloids have properties that are intermediate between those of metals and non-metals. They often *behave like semiconductors*.

For example: *Boron, Silicon, Germanium, Antimony, Arsenic, Tellurium.*



Noble Gases



Noble gases are *odorless, colorless, and very unreactive gases* found in Group 18 of the periodic table.

They rarely take part in chemical reactions due to their full outer electron shells.

For example: *Helium, Neon, Argon, Krypton, Xenon, and Radon.*



PHYSICAL PROPERTIES: METALS

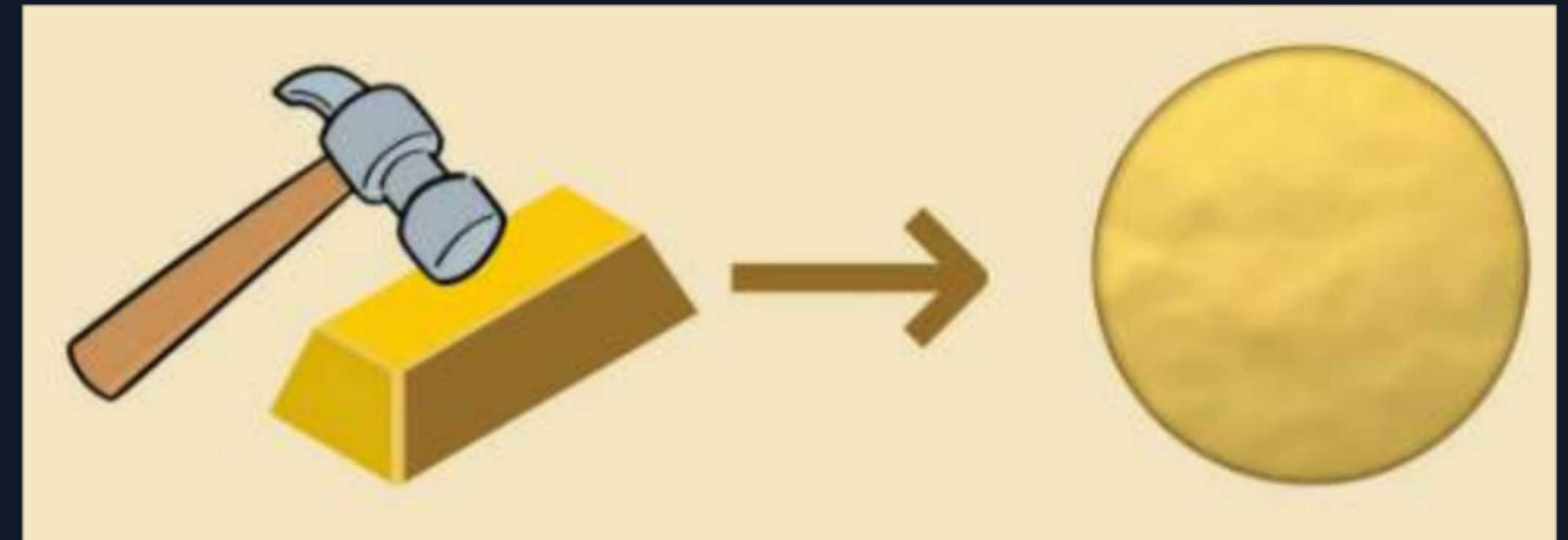


Malleability

✓ Metals possess the ability to be beaten into thin sheets without breaking. This property is known as malleability.

- **Examples:** Aluminium, Iron
- **Exceptions:** Zinc, Mercury

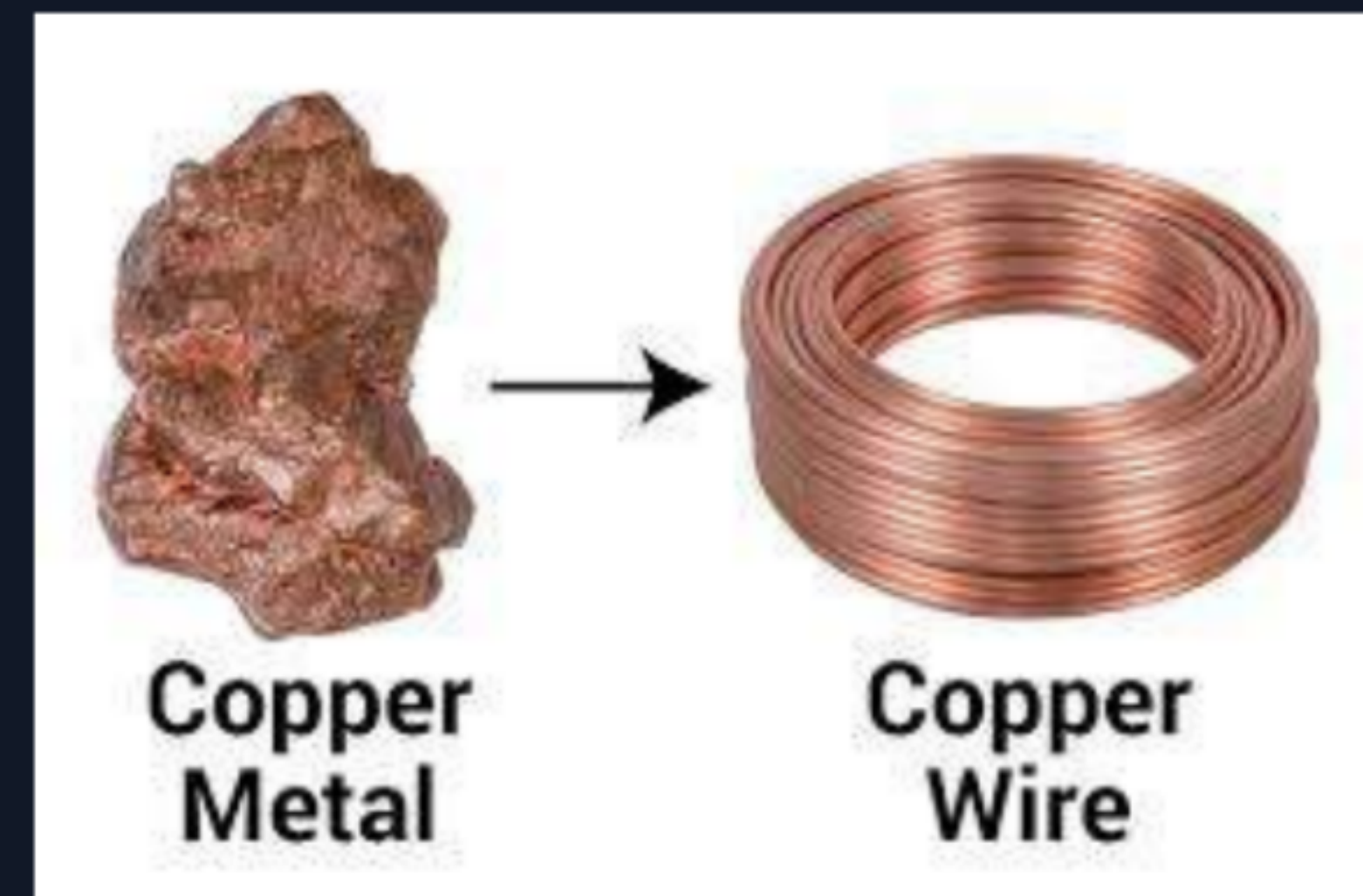
↓
Liquid



Ductility

✓ Metals can be drawn into thin wires without snapping. This feature is called ductility.

- **Examples:** *Copper, Aluminium*
- **Exceptions:** *Mercury, Zinc*



Hardness



Most metals are hard in nature, though the degree of hardness can vary among them.

- **Examples:** Aluminium, Iron
- **Exceptions:** Sodium, Potassium





Sonority

When struck, metals produce a ringing sound. This is referred to as sonority.

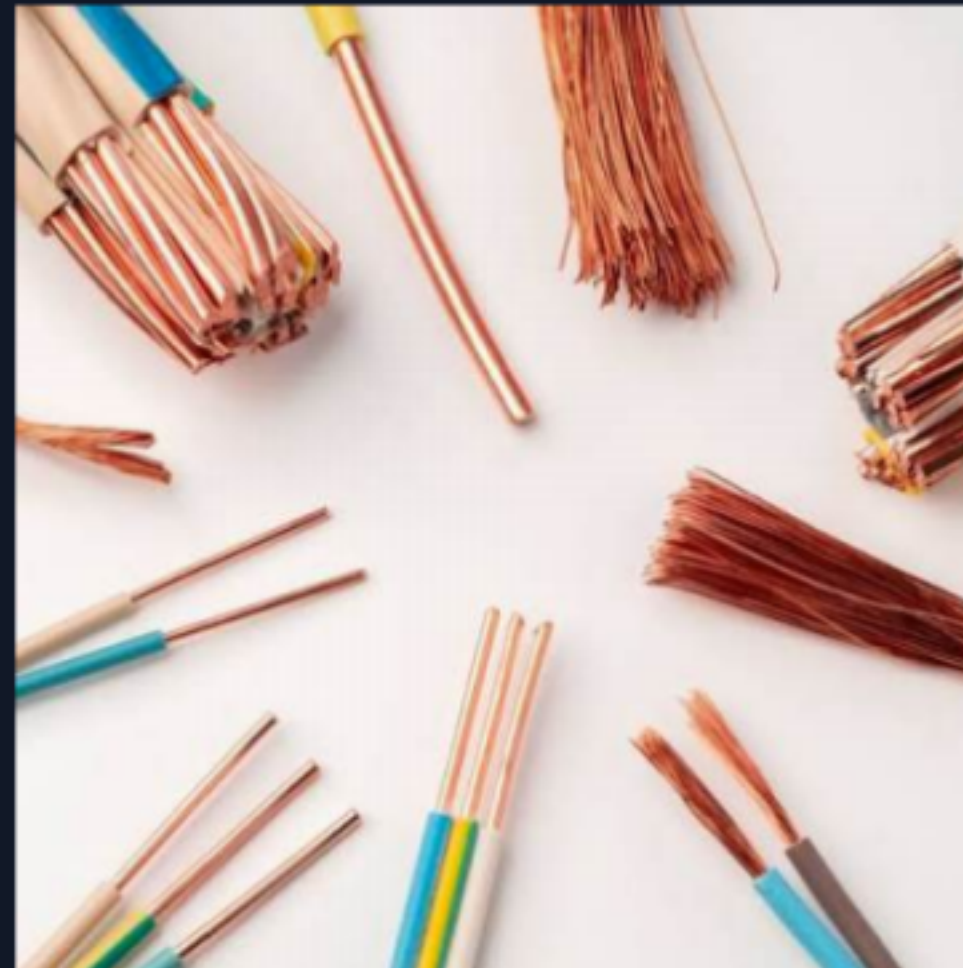
- **Examples:** Tin, Silver, Gold, Copper, Iron
- **Exceptions:** Mercury



Lustre

Metals have a shiny appearance and reflect light from their surfaces.
This shininess is termed as lustre.

- **Examples:** Copper, Aluminium, Gold *7 Silver*
- **Exceptions:** None



Melting and Boiling Point

Most metals melt and boil at high temperatures, usually above 1000°C .

Examples: Tungsten, Gold, Silver, Copper

Exceptions: ~~Sodium, Potassium, Mercury,~~



Electrical Conductivity

Metals are good conductors of electricity, hence they are used to make electrical wires.

- Examples: Silver, Copper
- Exceptions: Mercury, Lead



Heat Conductivity

Metals are good conductors of heat, which is why frying pans are made of metals.

- Examples: Silver, Copper
- Exceptions: Mercury, Lead

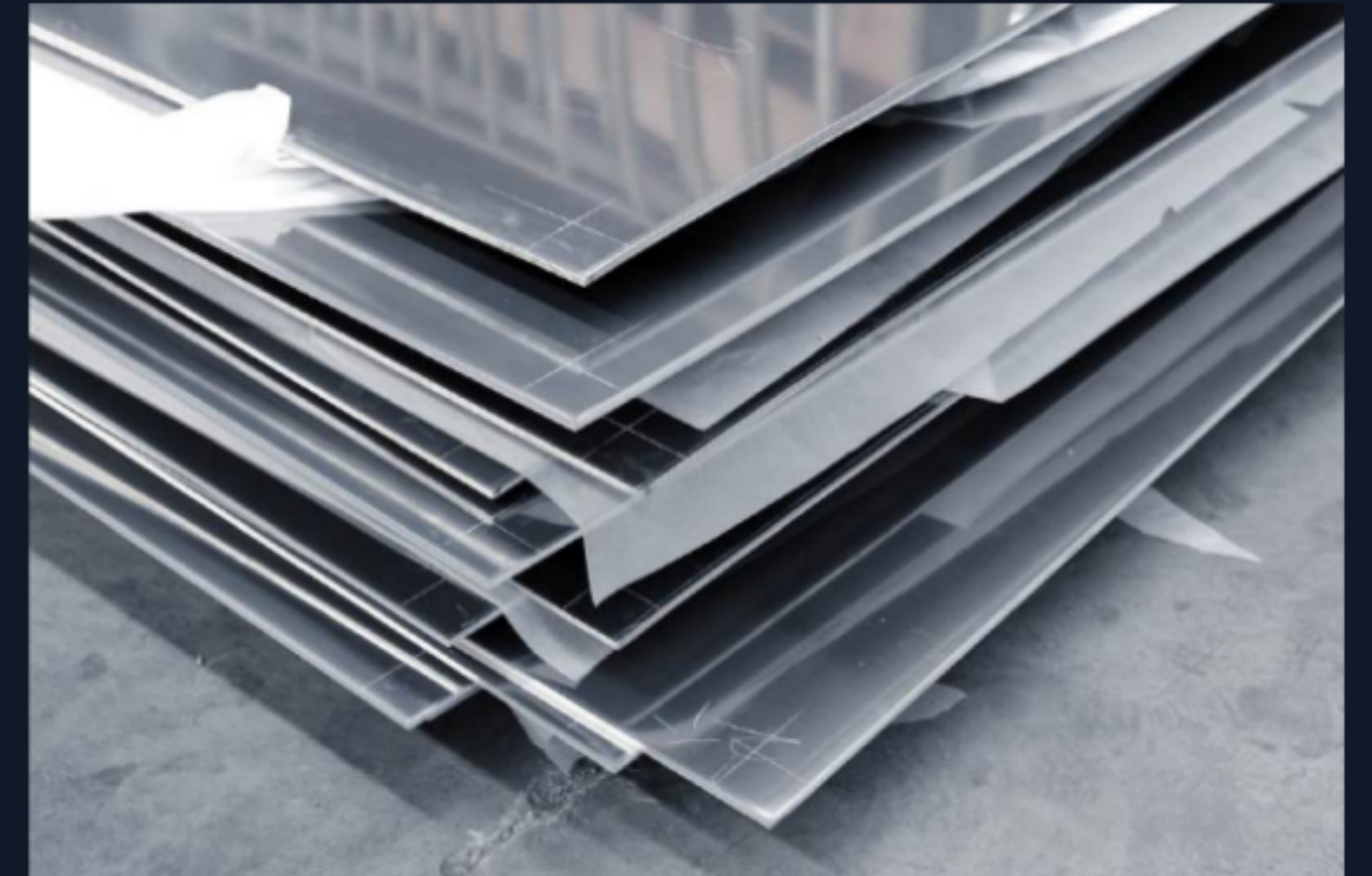


State at room Temperature

Under normal conditions, metals are usually in solid form.

Examples: *Aluminium, Iron*

Exception: *Mercury* (liquid at room temperature)



Metal Marvels

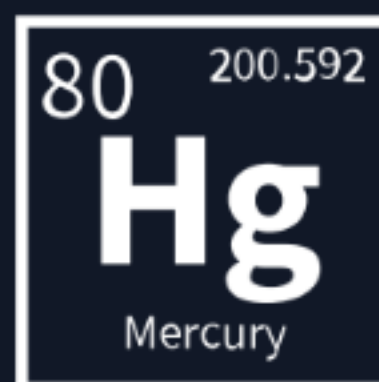
Gold:

The most malleable metal. It can be harmed into extremely thin sheets without breaking.



Mercury:

The only metal that exists in the liquid state at room temperature. It was traditionally used in thermometers



Platinum:

The most ductile metal. It can be drawn into very fine wires.

silver



Metal Marvels

Copper
Silver:
The best conductor of electricity.

④
[Gallium and Cesium]
have low melting points due to which they melt in the human palm.

Metals are hard except alkali metals like sodium, potassium, lithium, which can be cut with a knife.



PHYSICAL PROPERTIES: NON-METALS



Characteristics of Non-Metals

Non-metals are brittle and can break easily when hammered and cannot be beaten into thin sheets.

Example: Sulphur



Most non-metals do not conduct heat or electricity efficiently.

Example: Sulphur, phosphorus



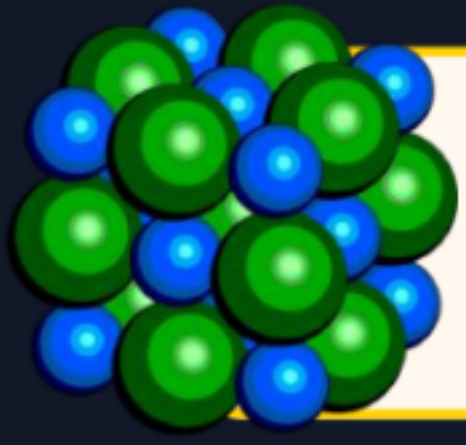
Found as solids, liquids and gases at room temperature.

for example:

carbon (solid),
Bromine (liquid),
oxygen (gas)

B_n





Characteristics of Non-Metals

Non metals are generally soft and less stronger than metals.

For example:
Sulphur

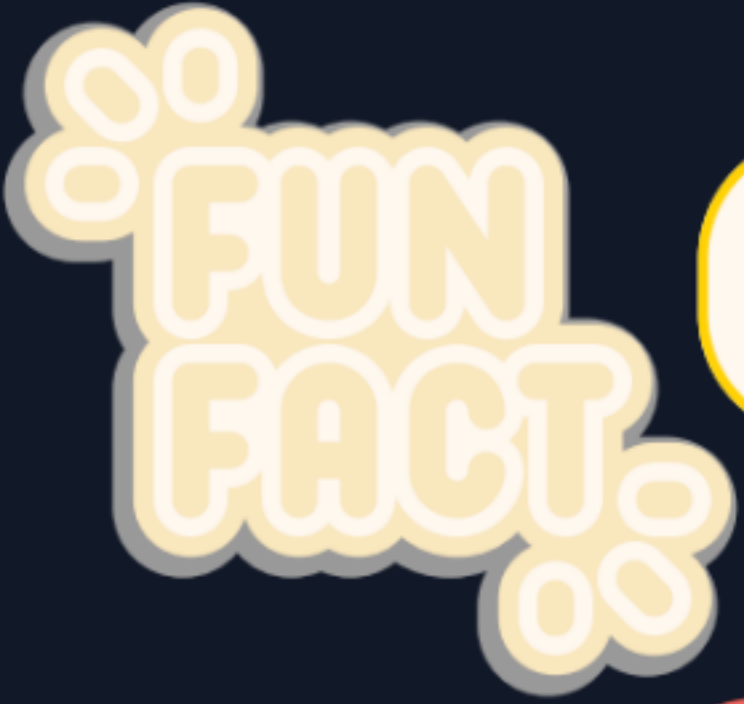


Non lustrous,
except iodine which is shiny



Generally have low melting and boiling points (except diamonds and graphite)





Science spotlights



Bromine:

The only non metals that exists as a liquid at room temperature

Diamond:

The hardest natural substance and excellent conductor of heat

Iodine and graphite:

The only non metals with lustrous (shiny) appearance



Graphite:

The only common non metal that conducts electricity.

QUES: Differentiate between Metals and non metals. [NCERT]

Property	Metals	Non-metals
<u>Physical State</u>	Generally solid at room temperature	Solids, liquids, or gases at room temperature
<u>Luster</u>	Lustrous (shiny)	Non-lustrous (dull) (except iodine)
<u>Malleability and Ductility</u>	Malleable and ductile	Brittle (delicate and easily broken)
<u>Conductivity</u>	Good conductors of heat and electricity	Bad conductors of heat and electricity (except graphite)
<u>Strength and Toughness</u>	Generally tough and strong	Neither tough nor strong
<u>Polishability</u>	Can be polished	Cannot be polished (except iodine)

Check yourself

Ques: The luster of a metal is due to

- a) It's high density.
- b) It's high polishing
- c) it's chemical inertness
- d) Presence of free electrons



Check yourself

Question: An unknown substance is dull, brittle, a poor conductor of electricity, and non-sonorous. It is most likely

- a) iron
- b) copper
- ☒ c) sulphur
- d) aluminum

→ Non-Metal



Check yourself

Ques: A metal can be converted into thin sheets but breaks when an attempt is made to draw it into a wire.

Which conclusion is most appropriate?

- ☒ a) The metal is malleable but not ductile.
- b) The metals is ductile but not malleable.
- c) The metals is brittle
- d) It is a non metal



Check yourself

Ques: You are given a hammer, a battery, a bulb wires and switch.

- a) How could you use the to distinguish between samples of metals and non metals.
- b) Asses the usefulness of these tests to distinguish between metals and non metals



Check yourself

Ques: Student A, graphite is a metal because it conducts electricity.
Student B: Mercury is a nonmetal because it is liquid.
Who is correct?

- a) only A
- b) Only b
- c) both a and b
- ☒ d) neither a nor b



Check yourself

Ques: Ram went to a temple and heard ringing of bells. He was surprised to hear the ringing sound and wondered why the temple bells are made of metal and not wood. Explain.

Answer: Metals have physical property of sonority which means that they can produce a ringing sound. When struck with a hard object. However, wood is not sonorous and doesn't produce a ringing sound. Therefore, bells in a temple are made of metal and not wood.



Check yourself

Ques: An engineer chooses aluminium for making aircraft bodies, copper for electrical wires and iron for bridges the property of each metal that makes it suitable for the given application.

High tensile Strength



Check yourself

Ques: A student identifies an unknown substance as a metal because it is shiny and Conducts electricity. Is this conclusion always reliable?

→ NO → Graphite
 ↓ ↓
 (S) (C)



Check yourself

Ques: Prema's friend bought sweets and chocolates to her place when she came to meet her. Prema saw that there was a layer of silver foil on top of the sweets, and the chocolates were wrapped in aluminum foil. She asked her mother how different metals could be used to make the foils. what would her mother explain to her

ans: Prema's mother explained that metals are malleable, which means they can be beaten into thin sheets. Due to this property, metals such as gold, silver, and aluminum can be converted into thin foils, which are used for decorating sweets and wrapping chocolates.



Check yourself

Q. Observe the following source carefully and answer the questions that follow:

1. Which nail will fall first?
2. In the above experiment, what should Riya keep the same in her activity to ensure a fair test?
3. (a) What would have happened if copper was replaced by lead?
4. Why are silver and copper good conductors of heat?

